NAME: ______________________ PERIOD: ________ DATE: ____________

Describing a Distribution

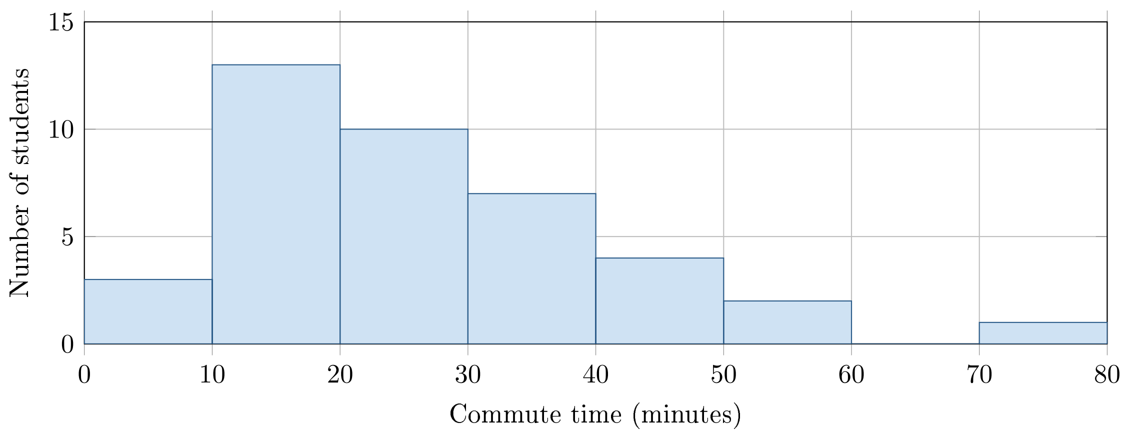
AP Statistics · Unit 1 · Topic 1.6 · B: 2026-09-17 · E: 2026-09-21

[STAR] TODAY'S PROBLEM

The histogram shows how long it took 40 students to get to school one morning, in minutes. Two students described the distribution.

Ana: "Roughly symmetric, centered near 20 minutes, spread from 5 to 45 minutes."

Ben: "Skewed right with a peak in the 10–20 minute interval; most students take under 30 minutes; one student at about 70 minutes is an outlier."



FIRST TAKE — before the video (5 min)

Whose description do you trust more, Ana's or Ben's? One sentence. No wrong answers yet.

[BOOK] DESCRIBING A DISTRIBUTION (keep on the page)

Describe **shape, center, variability, and unusual features** in context, with units. Center describes a typical response; variability describes how spread out values are. A longer right tail means skewed right; a longer left tail means skewed left. An outlier is unusually far from the rest. A gap has no observations. A histogram identifies intervals, not exact individual values.

Finish after the video:

DOK 1 **(a)** Which interval has the tallest bar?

DOK 2 **(b)** In two or three sentences, describe the commute times: shape, a typical value, overall spread, and one unusual feature. Use minutes.

DOK 3 **(c)** Whose description is better supported? Use two graph features to defend your choice. Explain why reporting only a typical commute time would leave a reader with an incomplete picture.

Frame: I trust ____________ because the graph shows ________________________ and ________________________.

Frame: A typical time alone would leave out ______________________________, which matters because ______________________________.

EXIT — turn this in

One thing the video changed about my first take: ____

Optional #1 — not collected

DOK 2 Sketch a dotplot of 15 quiz scores (0–10) that is skewed LEFT, unimodal, and has one low outlier. Then write the one sentence a reader would need in order to describe it in context.